Wind Direction Convention Error in the Downwind/Upwind Construction

Reproducible diagnostic for the bureaucrats and farms project

August 2026

Executive finding

The legacy pipeline combines two angular coordinate systems without converting between them. The wind variable is constructed with $\operatorname{atan2}(v, u)$, which is a mathematical angle measured counterclockwise from **East**. The geographic functions `geosphere::bearing()` and `geosphere::destPoint()` interpret their angles as compass bearings measured clockwise from **North**. Both values are expressed in degrees, but they are not directly comparable.

This is not a cosmetic labeling issue. In the second scenario below, the true wind points northwest and the AC representative point lies south-southeast of the grid, so the representative point is unambiguously upwind. The legacy code instead interprets the same numeric value as a southeast-pointing wind and classifies the representative point as downwind.

A real project example

The script searches the project shapefiles rather than inventing coordinates. It selects grid **116142** in the **SUNAM** assembly constituency (AC ID **449**), SANGRUR, Punjab. The grid centroid is 9.96 km from the AC representative point. The compass bearing from the grid centroid to that point is **164.986**°, effectively the requested 165° example.

For a mathematical angle α measured from East, the corresponding clockwise-from-North direction toward which the wind travels is

$$B_{\text{toward}} = (90^\circ - \alpha) \bmod 360^\circ.$$

The legacy code instead uses the numeric value α directly as if it were a compass bearing.

α from East	Correct compass	Legacy compass	AC bearing	Δ correct	Δ legacy	Correct	Legacy
55	35	55	165.0	130	110	Upwind	Upwind
135	315	135	165.0	150	30	Upwind	**Downwind (WRONG)**

Table 1: Angles and classifications. A point is downwind only when its circular angular separation from the wind-toward direction is less than 90°.

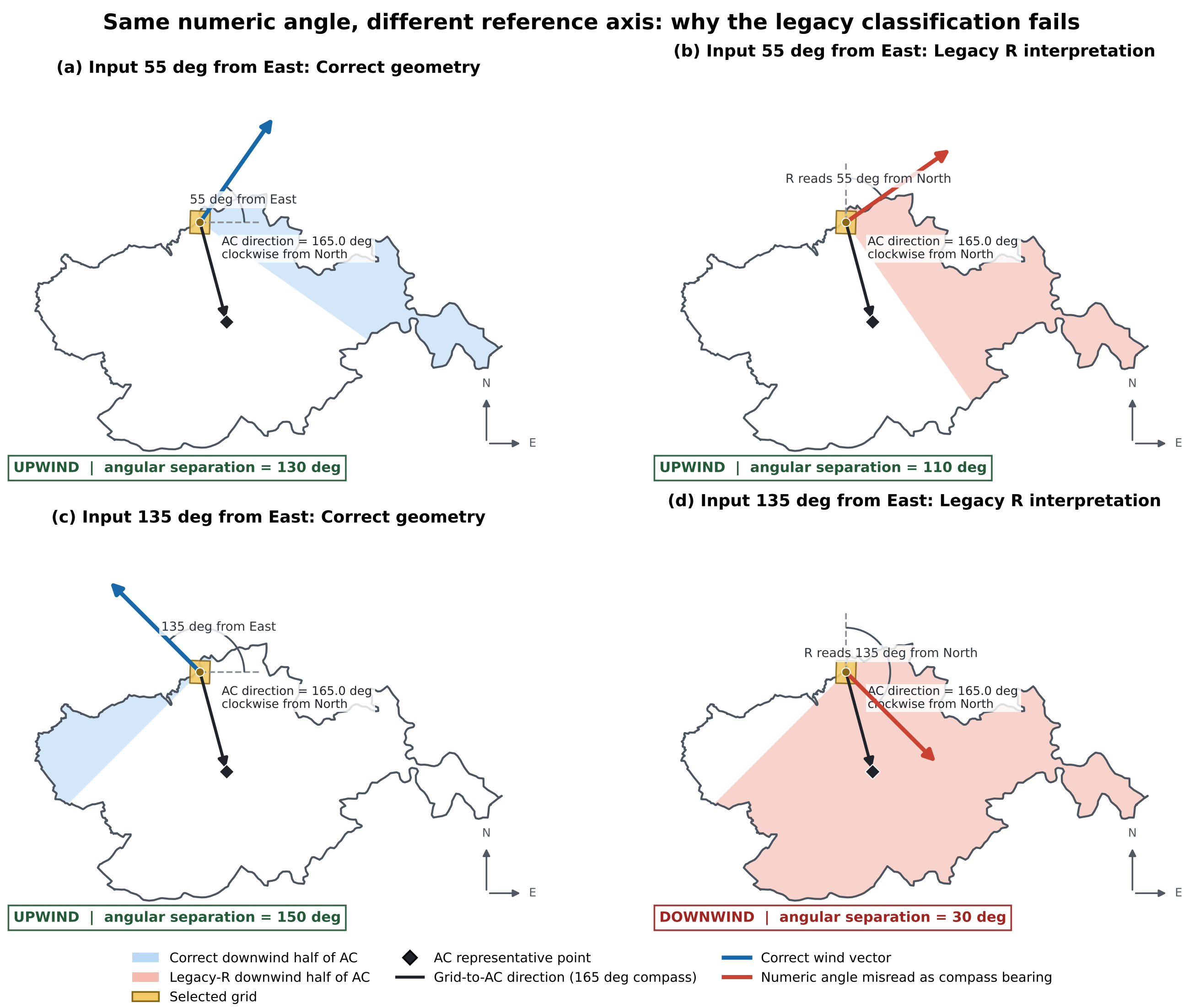

Figure 1: Four views of the same real AC/grid pair. Blue panels use the correct conversion from an East-referenced mathematical angle to a compass bearing. Red panels reproduce the legacy interpretation. The shaded portion is the AC half-plane labeled downwind; the black arrow points from the grid to the AC representative point. With a 55° input, both procedures call the point upwind, but only by chance. With a 135° input, the correct wind points northwest while the legacy interpretation points southeast, producing the incorrect downwind classification.

What is actually stored in `downup_ac_area`

The area indicator is defined as 1 when the area stored as downwind is larger than the area stored as upwind, and 0 otherwise. The diagnostic script reads the actual area parquet, selects the same real grid, and recomputes both half-plane intersections from the shapefile. Treating the raw wind number as a North-referenced compass bearing reproduces the stored areas to within 10^{-6} km^2. This proves that the convention error is present in the saved data, not only in a hypothetical reconstruction.

Period	Wind from East	Stored down	Stored up	Stored flag	Correct compass	Correct down	Correct up	Correct flag
2017-07	119.4	459.7	104.7	**1 (WRONG)**	330.6	24.0	540.4	**0**
2017-03	-75.0	159.7	404.7	**0 (WRONG)**	165.0	557.2	7.2	**1**

Table 2: Actual saved areas and classifications versus exact corrected intersections. Areas are in km^2; `downup_ac_area` equals 1 when downwind area is larger than upwind area.

For this selected grid, 136 of 148 nonmissing monthly observations change classification. The complete stored-by-corrected count is:

	Corrected 0	Corrected 1
Stored 0	12	112
Stored 1	24	0

These counts describe this one diagnostic grid only; they are not an estimate of the error rate in the full sample.

Why the negative angle is also misclassified

In March 2017, the stored raw direction is -74.971° from East. A negative Cartesian angle is valid: it points clockwise below the East axis. The correct compass-toward direction is $(90 - (-74.971)) \bmod 360 = 164.971°$. The legacy code instead normalizes the unconverted number to 285.029° and reads it from North. It therefore stores `downup_ac_area` $= 0$, whereas the corrected majority-area classification is 1.

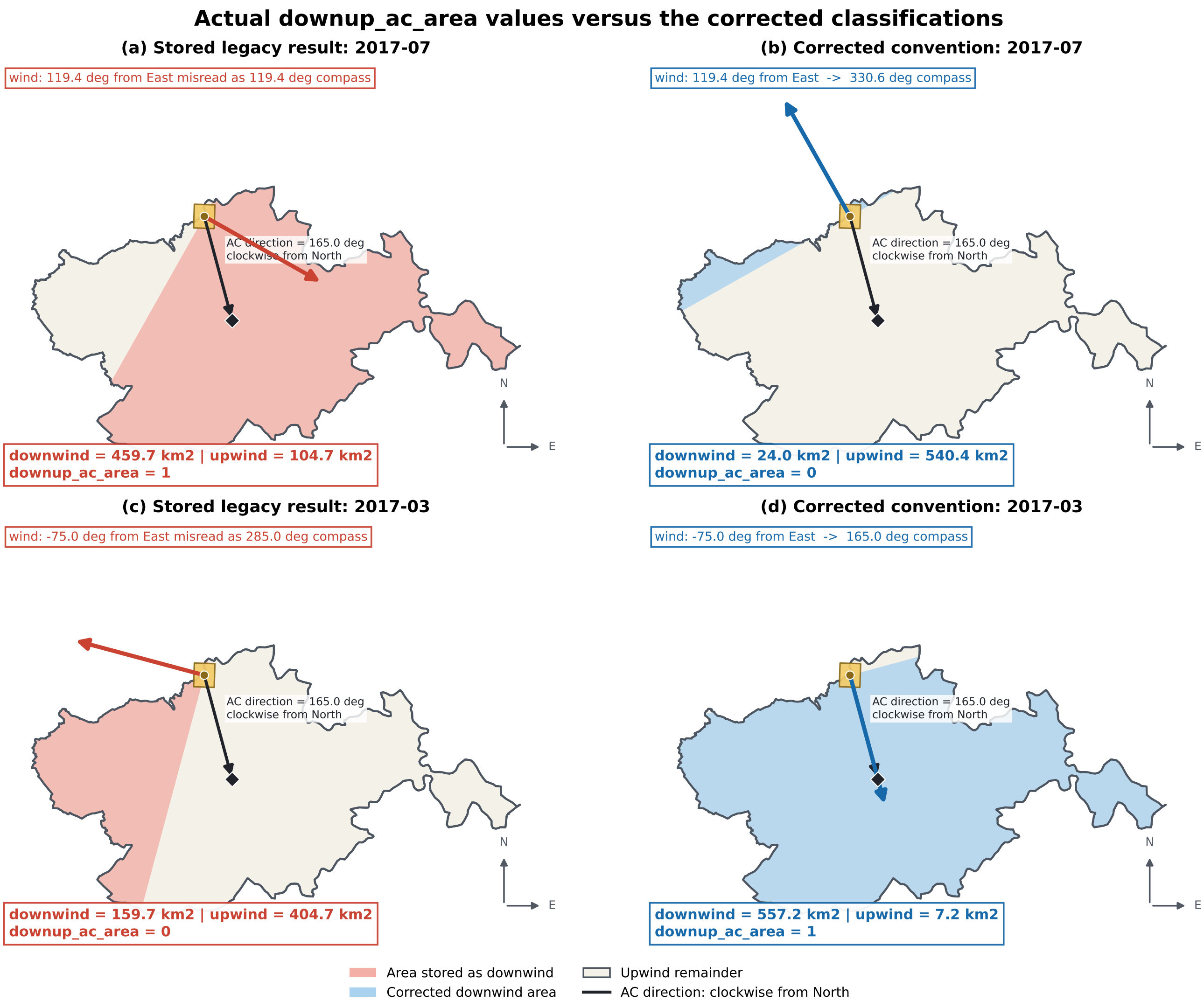

Figure 2: The values actually stored in the parquet (left column) versus the corrected calculations (right column). The first row is a positive wind angle; the second row is a negative wind angle. In every panel the black arrow to the AC representative point is a compass bearing measured clockwise from North. The wind input printed at the top is a mathematical angle measured from East. Red shading is the area stored as downwind; blue shading is the corrected downwind area. Both examples reverse `downup_ac_area`.

Corrected two-square area method

The corrected method first converts the wind angle to the compass-toward direction B. At the focal grid centroid p, define the unit wind vector $w = (\sin B, \cos B)$ and its perpendicular tangent $t = (\cos B, -\sin B)$. A large downwind square has corners

$$p - Rt, \quad p + Rt, \quad p + Rt + 2Rw, \quad p - Rt + 2Rw,$$

and the upwind square is constructed in the opposite direction, using $-w$. The radius R is chosen larger than the maximum distance from the grid centroid to the AC bounds. Therefore, the two squares cover the two complete halves of the AC and share only the perpendicular dividing line.

The exact areas are then

$$A_{down} = \text{area}(AC \cap S_{down}), \qquad A_{up} = \text{area}(AC \cap S_{up}), \qquad \texttt{downup\_ac\_area} = \mathbf{1}[A_{down} > A_{up}].$$

This construction is equivalent to clipping the AC by complementary half-planes, but showing the two finite squares makes the implementation and area accounting explicit.

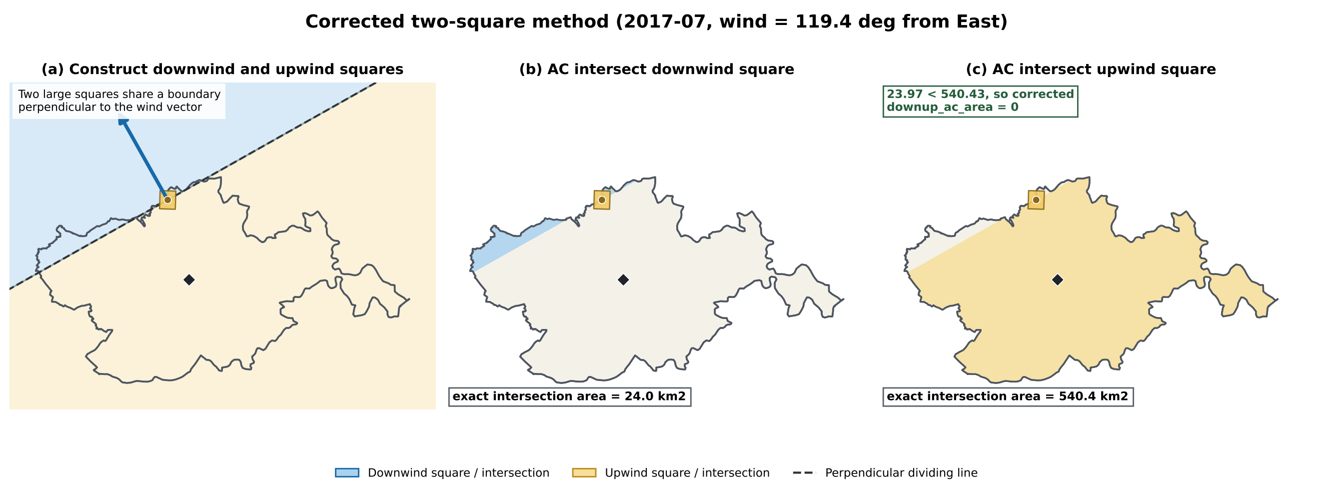

Figure 3: The corrected two-square construction for the July 2017 example. Panel (a) shows the complementary squares and their shared line, which is perpendicular to the corrected wind vector. Panels (b) and (c) show the exact AC intersections whose areas determine the binary classification.

Where the error enters

The wind-cleaning code calculates

```
wind_direction = atan2(v10, u10) * 180 / pi
```

This returns a Cartesian angle: 0° is East, 90° is North, 180° is West, and -90° is South. The downstream R code then uses

```
angle1 = wind_direction + 90
angle2 = wind_direction - 90
destPoint(grid_centroid, angle1, ...)
bearing_relative = bearing(grid_centroid, ac_point)
```

Here, 0° is North and 90° is East. The subsequent numeric comparison therefore mixes an East-referenced angle with a North-referenced bearing. In addition, ordinary inequalities do not correctly handle intervals that cross 0°/360°.

Required correction

The wind should first be represented as a compass direction toward which the air moves:

```
wind_direction_to = (atan2(u10, v10) * 180 / pi + 360) %% 360
```

The downwind test should use circular angular distance:

```
angle_difference = ((bearing_relative - wind_direction_to + 180) %% 360) - 180
downwind = as.integer(abs(angle_difference) < 90)
```

Monthly and rolling directions should also use circular means, preferably by aggregating vector components, rather than arithmetic means of degree values. For example, the arithmetic mean of 359° and 1° is 180°, while their circular mean is 0°.

Implications for existing outputs

- The population and area routines can agree with each other while both label the wrong geographical half-plane.
- Existing `downup_ac_pop` and `downup_ac_area` values should not be treated as directionally validated.
- Corrected wind directions require regenerating the wind panel, population measures, area measures, master dataset, and stacked data.
- Cardinal and diagonal unit tests should be retained so that East, North, West, South, 55°, and 135° cannot silently change convention again.

Source documentation

ECMWF defines positive u as west-to-east flow and positive v as south-to-north flow. See

`https://confluence.ecmwf.int/spaces/FCST/pages/111155337/`

ECMWF's discussion of meteorological wind angles and the special care required with `atan2` is available at

`https://confluence.ecmwf.int/pages/viewpage.action?pageId=226500971`